

4) $2 \cdot (1024)^3 = 2.147 \text{ GFlops} = \checkmark$ "work"

$2 \cdot 1024^2 \cdot 3 = 12,582,912 = Q$ "memory traffic"

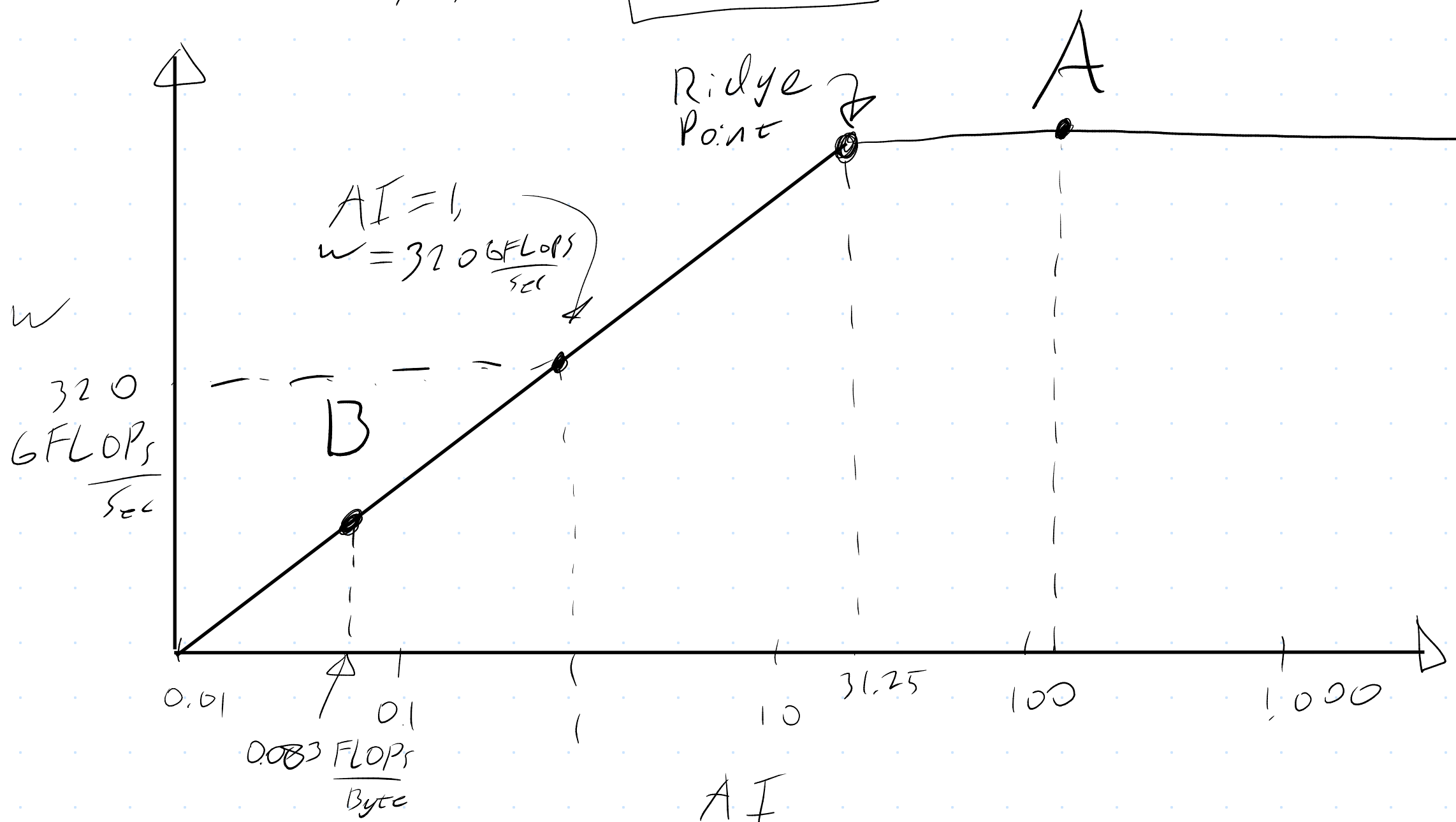
$AI = \frac{2.147 \text{ GFlops}}{12.582 \text{ MB}} = \boxed{170.67 \frac{\text{Flops}}{\text{Byte}}}$

B)

$w = 4,194,304 \text{ FLOPs}$

$Q = 3 \cdot 4,194,304 \left(4 \frac{\text{Bytes}}{\text{Flop}} \right) = 50,331,648 \text{ Bytes}$

$AI = \frac{50,331,648}{4,194,304} = \boxed{0.083 \frac{\text{Flops}}{\text{Byte}}}$



4) kernel A)

a) compute bound on this hardware

b) compute bound so maximum FLOPs

$31.25 \frac{\text{Flops}}{\text{Byte}} \cdot \frac{320 \text{ GB}}{\text{s}} = \boxed{10,000 \frac{\text{GFlops}}{\text{sec}}}$

c) Improve max compute because kernel A is compute-bound

kernel B)

a) memory bound on this hardware

b) memory bound so: $(320 \text{ GB/sec}) \cdot (0.083 \frac{\text{Flops}}{\text{Byte}}) = \boxed{26.56 \frac{\text{GFlops}}{\text{sec}}}$

c) Improve memory bandwidth / or add caching because kernel B is memory-bound